

Vishal Sinha

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EDUCATION

University of California, Irvine

Bachelor of Science in Computer Engineering

Irvine, CA

Sep 2024 – Jun 2028

EXPERIENCE

ArcelorMittal

Software Development Intern

Dubai, UAE

Jun. 2025 – Aug. 2025

- Built a full-stack web application (React, FastAPI, SQL Server) to track ~5,000 steel items moved daily
- Designed ERP-integrated REST APIs with optimized SQL queries for real-time item lookups
- Implemented immutable audit trails to support transactional logging and compliance requirements
- Digitized a paper-based logistics process, reducing manual tracking by ~10 hours per week

University Rover Challenge Team

Electrical Team Member

Irvine, CA

2026 – Present

- Contribute to the rover power subsystem architecture including battery integration and power distribution
- Assist in designing regulated power rails for Jetson compute modules, Pixhawk controllers, and onboard cameras
- Collaborate with mechanical and software teams to ensure reliable power delivery and electrical safety
- Participate in system testing and debugging of rover electronics and embedded hardware

PROJECTS

Spiderbot Quadruped Robot | *Embedded Systems, Arduino, C++*

2025 – Present

- Developing a quadruped spider robot using servo-based leg actuation and microcontroller control
- Implemented multi-servo control firmware for a quadruped robot using Arduino, mapping 12 joint servos in a 2D array structure and synchronizing GPIO-driven actuation to reset and calibrate leg positions
- Designed modular firmware architecture for motor control, gait sequencing, and sensor integration
- Testing locomotion stability and power usage across multiple walking algorithms

Battery Care System | *Arduino, Python*

Aug 2023 – Nov 2023

- Designed an automated battery management system using Arduino UNO R3 to disconnect charging cables at optimal battery levels
- Integrated relay modules, voltage sensors, and hardware control circuits for charging automation
- Implemented Python automation using `psutil`, `pyserial`, and `smtplib`
- Reduced overnight overcharging and improved long-term device battery lifespan

StudySwift – Study Timetable Generator | *Python*

Feb 2022

- Led a team project focused on improving student time management through automated schedule generation
- Developed timetable generation scripts using Python's `tabulate` library and text file parsing
- Implemented customizable user scheduling options for study sessions

TECHNICAL SKILLS

Programming: Python, C++, C, JavaScript, SQL

Embedded Systems: Arduino, Microcontrollers, Serial Communication, Sensors, Motor Control

Web: React, FastAPI, REST APIs

Tools: Git, Linux, SQL Server, Arduino IDE